

Ayman Gani

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EDUCATION

University of Waterloo

Bachelor of Software Engineering

Waterloo, ON

Sep. 2026 – Apr. 2031

EXPERIENCE

Firmware Team Member

UW Orbital

Sep. 2026 – Current

Waterloo, ON

- Completed onboarding designing mock temperature sensor driver and RTOS thermal management task
- Currently debugging non-deterministic unit test failures in CI/CD as first issue

Technology Lead

Reo, a Junior Achievement Company

Feb. 2026 – June 2026

Edmonton, AB

- Managed end-to-end tech infrastructure for a youth-led climate advocacy business
- Coded a custom interactive storefront on Shopify using HTML/CSS/JS, Liquid, and Anime.js
- Drove 1,000+ sessions and fulfilled \$800+ in orders while cutting termwise tech costs by 70% (\$50 to \$15)

Compiler Design and Optimization Lab Intern

University of Alberta

July 2025 – Aug. 2025

Edmonton, AB

- Audited computer architecture course materials, testing RISC-V Assembly labs such as CPU pipelining simulators
- Implemented assignment compiler passes targeting LLVM Intermediate Representation (IR)

Rated Contest Creator, Problem Setter

DM:OJ Modern Online Judge

Feb. 2025 – Apr. 2025

Edmonton, AB

- Created the Dr. Anne Anderson Coding Contest on Canada's top competitive programming site (185,000+ users)
- Authored 6 problems with 15+ test cases each, passing strict quality review to achieve rated status
- Managed 13 in-person and 300+ worldwide participants across 4,000+ code submissions

PROJECTS

Tutoring Club Automator | *Python, SciPy, Google Sheets*

Sep. 2025 – June 2026

- Built an automated tutor-tutee matching engine for a school tutoring club using linear sum assignment
- Reduced matching turnaround from 4+ days to under 2 hours, minimizing tedious spreadsheet sorting
- Deployed matching framework to another local tutoring organization, EduSpark, pairing 200+ total members

2026 CSDCMS CanSat Competition | *Python, Raspberry Pi, PyTorch, Jupyter*

Sep. 2025 – Apr. 2026

- Wrote onboard Python software for a can-sized sensor payload (temp/pressure/GPS/images) launched into the air
- Collected telemetry using I2C, UART and SPI, and ran a Monocular Height Estimation AI model onboard

Terminal Velocity | *C++, Make, Docker, GitHub*

June 2025 – Aug. 2025

- Engineered a 3D software rasterizer and space shooter from scratch, running at 15 FPS within a terminal emulator
- Processed 100+ game objects and 1,000+ triangle 3D model files with <5ms logic and rendering time per tick
- Architected custom input, audio, and frame-rendering pipelines designed for terminal emulator constraints

TECHNICAL SKILLS

Languages: C++, Python, C, JavaScript, HTML/CSS, C#, RISC-V Assembly, Java

Tools & Frameworks: Linux/Bash, Git, Make, Node.js, Unity, Docker, LLVM, LaTeX

Hardware & Protocols: I2C, SPI, UART, Raspberry Pi, PCB Layout (KiCad)

HONORS & AWARDS

Math, Sciences, and Technologies Award | *Edmonton Public School Board*

June 2026

Exceptional STEM Outreach Award | *CSDCMS CanSat Competition*

Apr. 2026

3rd Place | *University of Alberta Programming Contest*

Mar. 2025